



Description

Done to carry out a study of the deformation of the seat due to the load applied to it.

Simulation of FEA

Date: 21 October 2023
Designer: Group 4
Study name: Seat Buckling
Analysis type: Static

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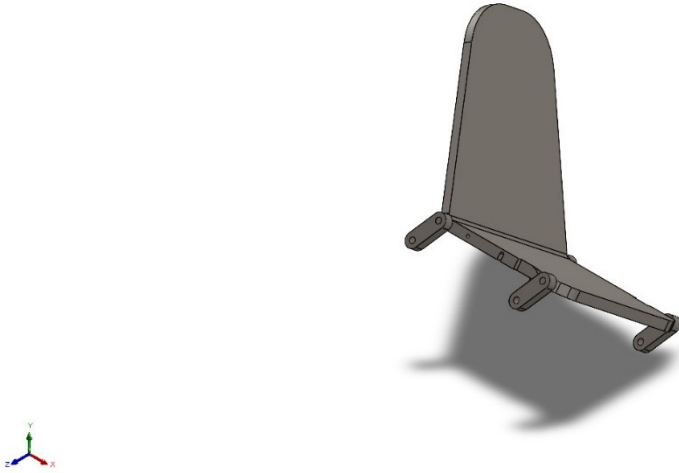
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Assumptions

The backrest and the bottom of the pins is fixed.

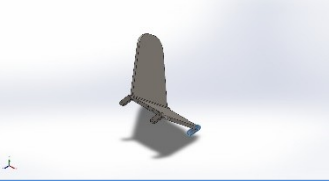
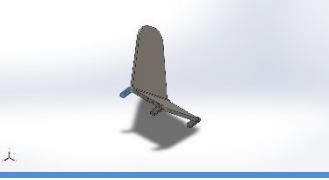
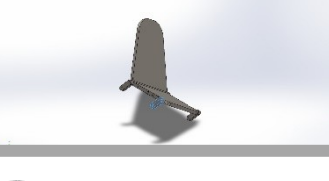




Model name: FEA

Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Boss-Extrude1 	Solid Body	Mass:0.351222 kg Volume:4.56133e-05 m ³ Density:7,700 kg/m ³ Weight:3.44198 N	C:\Users\Ibrahim\Downloads\ assemb_final\ link.SLDPRT Oct 20 12:04:38 2023
Boss-Extrude1 	Solid Body	Mass:0.351222 kg Volume:4.56133e-05 m ³ Density:7,700 kg/m ³ Weight:3.44198 N	C:\Users\Ibrahim\Downloads\ assemb_final\ link.SLDPRT Oct 20 12:04:38 2023
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Study Properties

Study name	Static 2
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\Ibrahim\Downloads\assemb_final)

Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

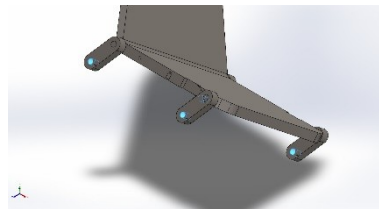
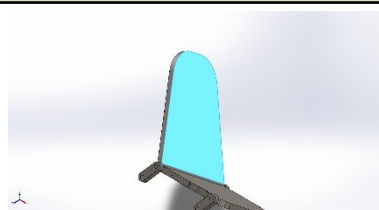
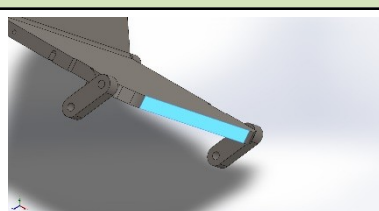


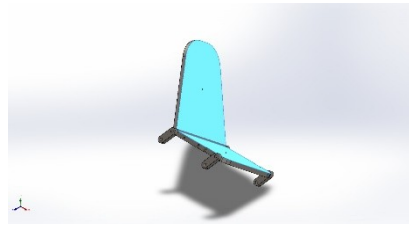
Material Properties

Model Reference	Properties	Components
	Name: Alloy Steel (SS) Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 6.20422e+08 N/m² Tensile strength: 7.23826e+08 N/m² Elastic modulus: 2.1e+11 N/m² Poisson's ratio: 0.28 Mass density: 7,700 kg/m³ Shear modulus: 7.9e+10 N/m² Thermal expansion coefficient: 1.3e-05 /Kelvin	SolidBody 1(Boss-Extrude1)(link-1), SolidBody 1(Boss-Extrude1)(link-2), SolidBody 1(Boss-Extrude1)(link-3), SolidBody 1(Boss-Extrude1)(link-4), SolidBody 1(Cut-Extrude7)(seat-bottom-1)
Curve Data:N/A		

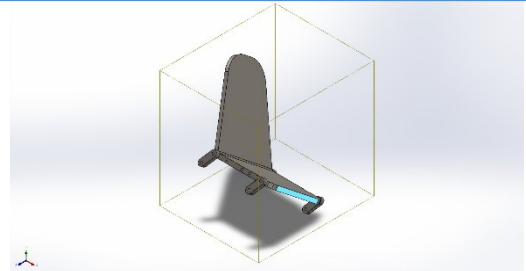


Loads and Fixtures

Fixture name	Fixture Image	Fixture Details			
Fixed-3		Entities: 4 face(s) Type: Fixed Geometry			
Resultant Forces					
Components	X	Y	Z	Resultant	
Reaction force(N)	28.4735	54.4326	-43.1942	75.0958	
Reaction Moment(N.m)	0	0	0	0	
Fixed-7		Entities: 1 face(s) Type: Fixed Geometry			
Resultant Forces					
Components	X	Y	Z	Resultant	
Reaction force(N)	491.224	263.435	-207.618	594.814	
Reaction Moment(N.m)	0	0	0	0	
Fixed-8		Entities: 1 face(s) Type: Fixed Geometry			
Resultant Forces					
Components	X	Y	Z	Resultant	
Reaction force(N)	-172.641	113.547	-89.7429	225.281	
Reaction Moment(N.m)	0	0	0	0	

Load name	Load Image	Load Details
Force-1		Entities: 2 face(s) Type: Apply normal force Value: 750 N

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh



Mesh information

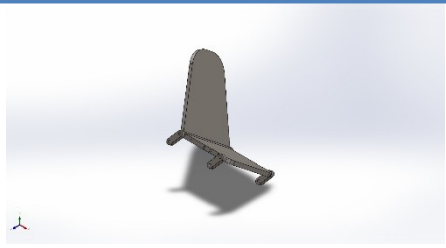
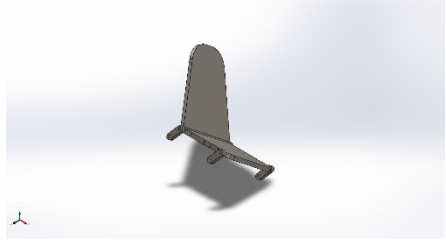
Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	31.9239 mm
Minimum element size	2.2961 mm
Mesh Quality	High
Remesh failed parts independently	Off

Mesh information - Details

Total Nodes	32155
Total Elements	17660
Maximum Aspect Ratio	24.265
% of elements with Aspect Ratio < 3	82.5
Percentage of elements with Aspect Ratio > 10	0.708
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:08
Computer name:	IBRAHIM-PC



Sensor Details

Sensor name	Location	Sensor Details
Mass1		Value : 4,064.61 g Entities :5 component(s) Result :Displacement Component :URES: Resultant Displacement Criterion :Model Max Step Criterion : Across all Steps Step No.:1 Alert Value: NA
Displacement1		Value : 9.305e-04 mm Entities : Result :Displacement Component :URES: Resultant Displacement Criterion :Model Max Step Criterion : Across all Steps Step No.:1 Alert Value: NA

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	347.056	431.414	-340.555	650.034

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

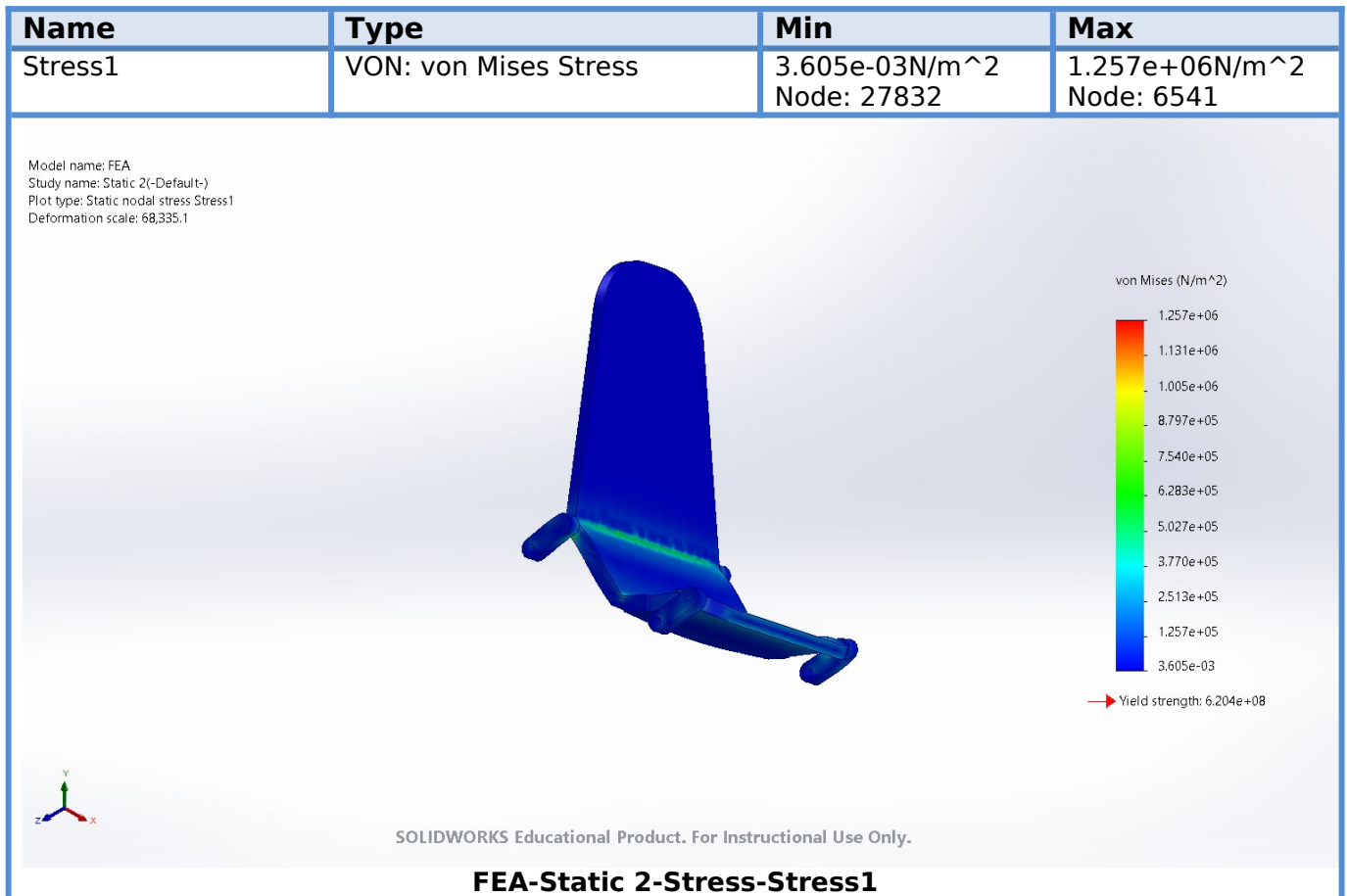
Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.0229861	-0.0706891	0.038059	0.0835093

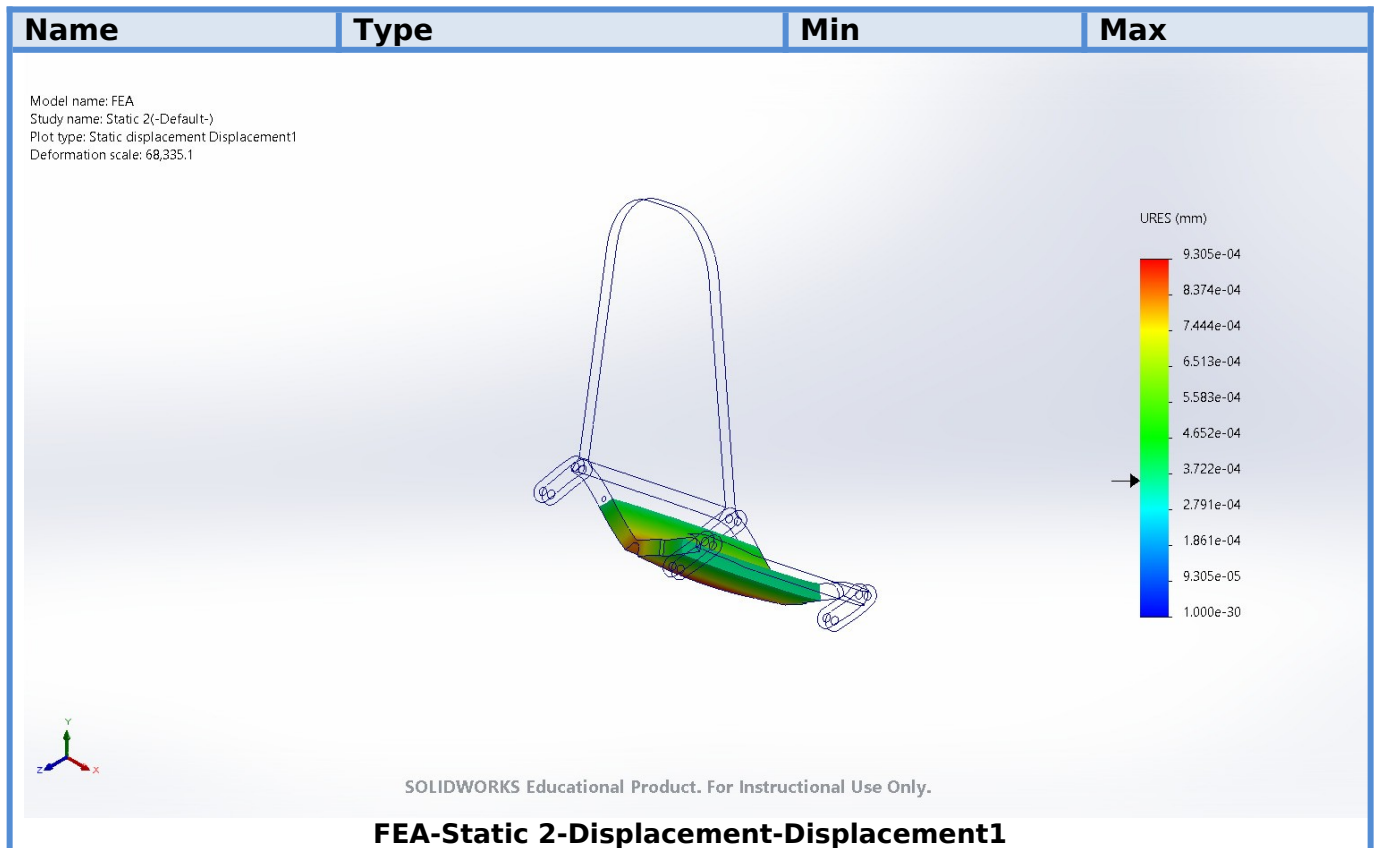
Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

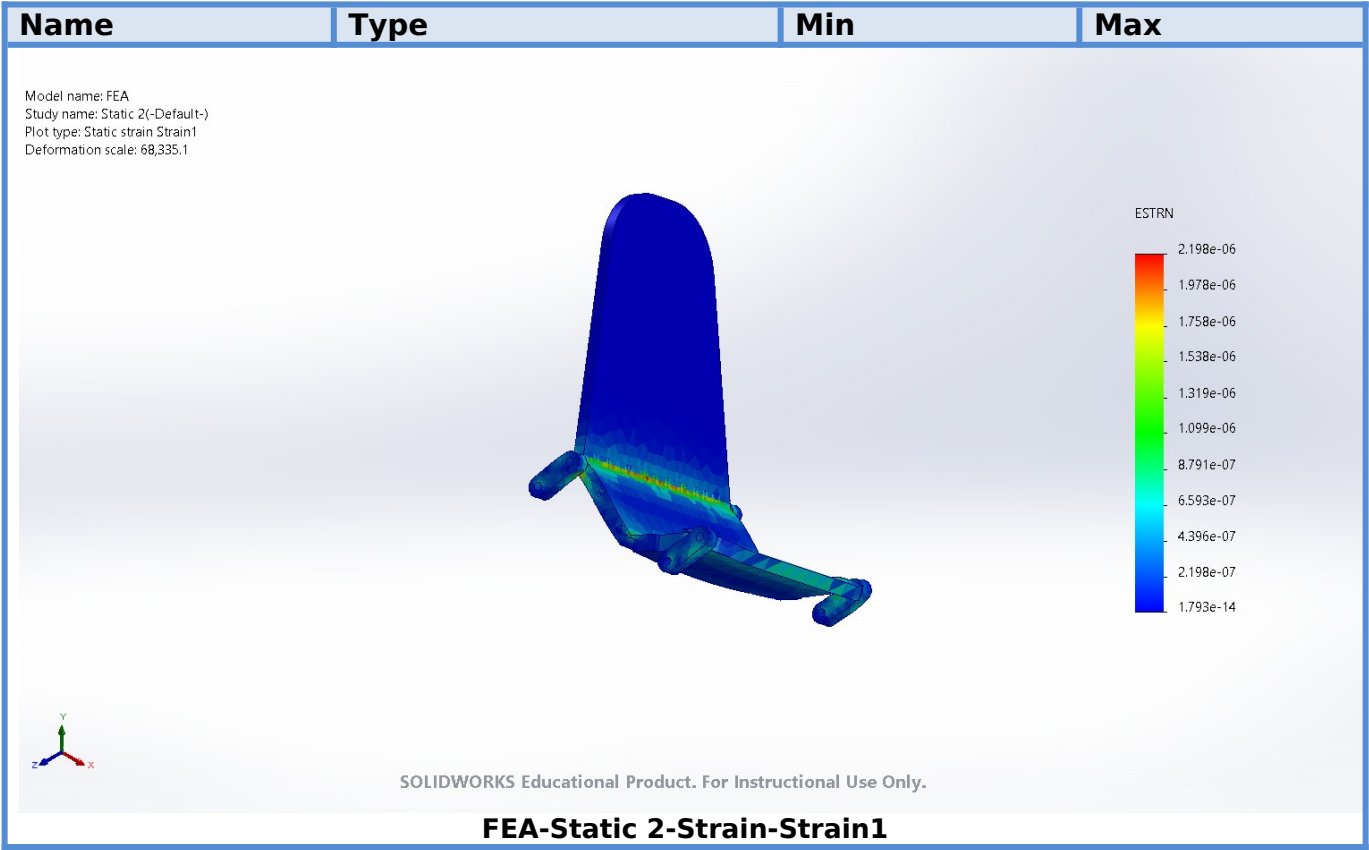
Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 7	9.305e-04mm Node: 31273



Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.793e-14 Element: 15157	2.198e-06 Element: 14937



Name	Type
Displacement1{1}	Deformed shape

